

Research Diagnosis: Evaluation of LLMs on Bloom’s Taxonomy (Monash CLO)

Prepared for: Muhammad Daniyal
Date of literature scan: 21 April 2026
Scope: 2025–2026 evidence only (Anderson & Krathwohl 2001 and Li et al. 2022 cited as anchors)

0. One-line verdict

What you have built is a **very thorough benchmarking study** — probably the largest single-author comparison on the Monash CLO dataset to date — but in its current form it is a **technical report, not a research contribution**. You have measured; you have not yet *claimed* anything new. Your own results reproduce the 2022 Monash ceiling (~93%) rather than breaking it, and the methodological questions you pose have already been answered (sometimes better, sometimes not) in four 2025 papers. The fix is not to run more models; it is to reframe the study around a problem the field has *not yet solved*. I give concrete proposals in §5.

1. What you actually did (reconstructed from your slides)

Stage	Models	Setup
EDA	—	21,379 → ~18,773 rows, Understand/Apply dominate, Remember rare (886), ~11% multi-label
Classical ML	MNB, CNB, LR, SGD-Huber, Linear SVM (word), Linear SVM (char), KNN+LSA, Decision Tree, Random Forest, Extra Trees, XGBoost (LSA-200), LightGBM (LSA-200), HistGB (LSA-200)	TF-IDF / LSA-200 features
Deep Learning	LSTM, BiLSTM, TextCNN, BiLSTM+Self-Attention, DistilBERT (fine-tuned)	75/25 split + 10-fold

Stage	Models	Setup
Ensembles	Soft-voting (LR+SVM+CNB), Soft-voting (LR+SVM+BiLSTM+CNN), Stacking (LR+SVM+CNB+RF → LR)	Model-level aggregation
LLM zero-shot	Mistral-7B v0.2, Llama-3/3.1-8B, Phi-3.5/Phi-4-mini, Qwen2-7B, Qwen2.5-{1.5B, 3B, 7B}, SmoLLM2-1.7B, Gemma-2-{2B, 9B}, Mistral-Nemo	6-class prompt
LLM few-shot	same 13 (minus one)	same prompt + exemplars
QLoRA fine-tune	Mistral-7B-Instruct-v0.3, Qwen2-7B-Instruct, Gemma-2-9B-IT	4-bit NF4, r=16, 3 epochs, ~16K rows

Headline numbers. DistilBERT fine-tuned: macro-F1 **0.918**, accuracy **92.7%**. Mistral-7B QLoRA: accuracy **93.22%**, weighted-F1 **0.932**, macro-F1 **0.921**. Qwen2-7B QLoRA: **92.47%**. Gemma-2-9B QLoRA: **20.24%** (collapsed — a chat-template bug, not a capability failure).

2. Is this progressive research? — Honest answer: not yet.

You asked me to be direct, so here it is. Three structural problems:

2.1. No research question. “We compared 20+ models” is a leaderboard, not a contribution. Every one of your 20 models is known to work on text classification. The field already knows BERT beats TF-IDF, that transformers beat LSTMs, and that LLMs with few-shot beat LLMs with zero-shot. Replicating this on one more dataset is a class project, not an EDM/AIED/LAK paper.

2.2. No ceiling broken, no bound established. Li et al. 2022 reported Cohen’s κ up to 0.93 and F1 up to 0.95 on the same 21,380-row dataset with BERT. Your DistilBERT gives macro-F1 0.918 and your QLoRA-Mistral gives 0.921. You are at — not past — the 2022 state of the art. The June 2025 MDPI *Electronics* paper (Bayesian-optimized ensemble with DistilBERT+TF-IDF) already observed “the highest accuracy reported in existing research has not exceeded 93.5%” and positioned itself explicitly against that ceiling. Anyone reviewing your paper will ask: *what’s new?*

2.3. The questions you answer have already been answered in 2025. Rapid-fire summary of what 2025 literature already establishes, which your experiments largely re-demonstrate:

Finding in your slides	Already shown in 2025
Few-shot > zero-shot for LLMs on Bloom	Almatrafi & Johri 2025 (<i>Computers & Education: AI</i>) — prompt engineering strategies on GPT-4 for CLOs
Small (<3B) LLMs fail even with exemplars	Scaria et al. AIED 2024 + replicated in 5 subsequent 2025 papers
Higher-order levels (Analyze/Evaluate/Create) are the hard cases	Huber & Niklaus COLING 2025; Ma et al. 2025 (BloomAPR)
Fine-tuned BERT variants still beat prompted LLMs	Almatrafi & Johri 2025 (explicitly); Gulwani arXiv 2511.10903 (Nov 2025)
QLoRA lets 7–9B models reach BERT-class accuracy	Well established since Dettmers et al. 2023; routine in 2025
Gemma tokeniser/chat-template mismatch hurts fine-tuning	Documented repeatedly in 2025 Gemma-2 fine-tune reports

None of these are wrong observations. They are *confirmatory*. Confirmation has value, but it is a ~3-page short paper, not a thesis-scale contribution.

3. How your results compare to 2025–2026 literature (concrete numbers)

3.1 Direct comparators on CLO / learning-objective classification

Study	Dataset	Best model	Best score
Li, Rakovic, Poh, Gašević, Chen — EDM 2022 (<i>Monash dataset origin</i>)	21,380 Monash CLOs	BERT (binary per-class)	Cohen’s $\kappa \leq \mathbf{0.93}$, F1 $\leq \mathbf{0.95}$
Almatrafi & Johri — <i>Computers & Education: AI</i> , 2025	1,000 annotated LOs	BERT fine-tuned (beat every GPT-4 prompt strategy)	not directly comparable; ran CoT, rhetorical-situation, multiple-binary prompts

Study	Dataset	Best model	Best score
Ghannam et al. — MDPI <i>Electronics</i> , June 2025 (Bayesian-optimized ensemble, TF-IDF + DistilBERT features)	Yahya question bank	Soft-vote RF+SVM+GB+XGB, Bayesian-tuned	Accuracy >93.5% (explicitly claims to break the prior ceiling)
Gulwani & Kothari — arXiv 2511.10903, Nov 2025	600 sentences, 6 Bloom levels	SVM + synonym aug	94% accuracy (but on 600 sentences, not 18K)
Your study (2026)	Monash 18,773	DistilBERT / QLoRA-Mistral	macro-F1 0.918 / 0.921 , acc 92.7 / 93.2%

Observation: your macro-F1 on the full Monash split is **essentially identical** to what Li et al. already reported in 2022 with vanilla BERT. You did not break the ceiling; you reached it again with more models.

3.2 Adjacent benchmarks with Bloom as the organising axis (2025–2026)

These are the papers your related-work section is going to have to engage with, and which define where the field has moved since 2022:

- **Huber & Niklaus — COLING 2025** (“LLMs meet Bloom’s Taxonomy”): shifted the question from “classify LOs” to “do LLM benchmarks cover the Bloom hierarchy?” Their finding — Evaluate/Create are severely under-represented in MMLU, BBH, AGIEval — is the most-cited 2025 Bloom-LLM paper and will be in every related-work section from now on.
- **Ma et al. — BloomAPR, arXiv Sep 2025**: uses Bloom levels as a *dynamic* benchmark axis for LLM-powered program repair; explicitly shows performance drops across Remember → Analyze (81.57% → 13.46%).
- **Zhou et al. — MedBloomEval, 2025**: Bloom levels as a reasoning-depth axis for medical LLMs.
- **Zoumpoulidi et al. — BloomWise, MathNLP 2025**: uses Bloom levels as a *prompting scaffold* for math reasoning, with an early-stopping criterion when consecutive levels agree.
- **Huber & Niklaus’s linear-probing follow-up (arXiv 2602.17229, Feb 2026)**: mechanistically probes transformer layers for Bloom-level information — finds errors are structured, adjacent-level confusions dominate.
- **"Automated Benchmark Generation from Domain Guidelines" (arXiv 2601.20253, Jan 2026)**: auto-generates Bloom-aligned MCQs from expert guidelines for teaching/dietetics/caregiving; finds LLMs sometimes do *better* on Analyze than on Remember — the opposite of

what Huber & Niklaus saw.

- **"Do Existing Benchmarks Accurately Evaluate LLMs?" – Findings of EMNLP 2025** (arXiv 2511): quantifies cognitive-level coverage in 20+ LLM benchmarks using Shannon–Wiener, Simpson, JSD, Gini over Bloom levels.

The field has moved from “*can we classify LOs into Bloom levels*” (~2020–2022, where you are) to “*what can Bloom tell us about LLMs*” (2025–2026, where the frontier is). You are asking a 2022 question in 2026.

3.3 Where your numbers *do* differ from the literature (real findings worth rescuing)

Don’t throw everything out. Three genuinely interesting signals in your results that are *not* already in the 2025 literature:

1. **Qwen2.5-7B beats Gemma-2-9B in zero/few-shot** but Gemma-2-9B beats Qwen2-7B on few-shot accuracy (67.5% → 72% for Gemma vs. 61% → 75% for Qwen-2.5). This *parameter-count-vs-few-shot-gain* interaction, at 7–9B scale, is not in the 2025 literature I can find.
2. **Fine-tuned 7B Mistral (0.921 macro-F1) ≈ fine-tuned DistilBERT (0.918)** on the same split with QLoRA. Almatrafi & Johri 2025 found BERT beats GPT-4 prompt strategies, but they never fine-tuned a 7B open model. Your observation — that QLoRA at ~40M trainable params closes the gap — is a genuinely fresh data point for the CLO task specifically.
3. **Few-shot hurts Qwen2.5-1.5B (45.7% → 30.5%)** — a *negative* interaction between exemplars and very small models, in the same family that benefits at 7B. This is the kind of result that *small-model researchers* would actually read.

These are the seeds. Everything else is either a leaderboard or confirmation.

4. Direct conflicts between your results and recent literature

Claim in 2025 lit	Your result	Conflict or not?
Huber & Niklaus 2025: LLMs do better on lower Bloom levels	Your Qwen2.5-7B <i>gains most</i> on Remember (rarest class) in few-shot (+23pp)	Nuance, not conflict. Their claim is about raw ability; yours is about exemplar-induced gains. Worth articulating.

Claim in 2025 lit	Your result	Conflict or not?
Gulwani 2025 (arXiv 2511.10903): OpenAI/ Gemini ≈ 0.72 – 0.73 acc zero-shot, beats open LLMs	Qwen2.5-7B hits 0.61 ZS, 0.75 FS; Gemma-2-9B 0.675 ZS, 0.72 FS	No real conflict — different dataset (600 sentences vs 18K), but you could claim that <i>at 7–9B</i> , open models now match GPT-4o-mini zero-shot, which is a useful point.
MDPI 2025 Bayesian ensemble: $>93.5\%$ is achievable	Your best is 93.2%	You are below. They beat you.
Almatrafi & Johri 2025: fine-tuned BERT $>$ best-prompted GPT-4	DistilBERT 0.918 $\approx$ QLoRA-Mistral 0.921 $>$ any of your prompted LLMs (best FS ≈ 0.75 acc)	Confirms their finding, does not challenge it.
BloomAPR 2025: LLM performance cliffs at Analyze layer ($\sim 43\%$ Apply $\rightarrow \sim 13\%$ Analyze on perturbed data)	DistilBERT confusion: Analyze recall 0.91, Evaluate 0.91 — no cliff	Genuine conflict on surface , but they evaluate reasoning, not text classification. Don't overclaim.

5. Your three questions, answered directly

5.1 “How different are my results from previous studies?”

Short answer: barely. On the Monash dataset you match Li et al. 2022 (BERT $\kappa=0.93$, $F1=0.95 \rightarrow$ you get macro-F1 0.918–0.921). On the broader Bloom-NLP literature, you are in the same 72–94% accuracy band as every 2024–2025 study, with no clear win. The three *genuinely differentiated* findings are listed in §3.3. If you write the paper as it stands, a reviewer will write: “*The authors confirm known results with a larger model zoo. Contribution is incremental.*”

5.2 “What novelty / new methodology should I bring?”

Here are **five candidate contributions**, ranked by novelty-defensibility-cost trade-off. Each cites why the 2024–2026 literature has left it open.

Option A – Hierarchy-aware / ordinal classification for Bloom ★ strongest defense

Bloom's six levels are *ordinal* (Remember < Understand < Apply < Analyze < Evaluate < Create), but every single paper I found — Li et al. 2022, Almatrafi & Johri 2025, MDPI *Electronics* 2025, Gulwani 2025, your study — trains with **nominal cross-entropy**. Misclassifying Create as Remember is penalised the same as misclassifying Create as Evaluate. This is obviously wrong pedagogically and wrong statistically.

Why it's defensible (2025–2026 evidence):

- The BloomBERT 2025 GitHub README explicitly calls this out as open future work (“ordinal regression formulation could enforce such constraints”).
- Huber & Niklaus's Feb 2026 linear-probing paper (arXiv 2602.17229) shows empirically that “*misclassifications predominantly occur between adjacent Bloom levels*” — the error structure is ordinal, but the loss is not.
- The ordinal-loss literature is mature (CORAL, CORN, weighted- κ loss, class-distance-weighted CE — Polat et al. 2025 *Expert Systems with Applications*; Shi, Cao & Raschka 2023; Bérchez-Moreno et al. *Neurocomputing* 2025 “dlordinal” Python package).
- In medical imaging 2025–2026, swapping CE for ordinal-aware loss gives consistent macro-F1 gains on BI-RADS and lumbar CT (both *Diagnostics / Journal of Clinical Medicine*, late-2025 MDPI).

What you'd actually do: fine-tune DistilBERT and/or Mistral-7B QLoRA with (a) weighted- κ loss, (b) CORAL-style rank-consistent output heads, © class-distance-weighted cross-entropy. Report not just macro-F1 but **quadratic weighted kappa (QWK)** and **mean absolute error in Bloom levels** — metrics the current Bloom literature *does not report*. Show that models look ~equivalent on nominal F1 but differ meaningfully on QWK.

Why nobody has done this for Bloom specifically: because CLO classification grew out of educational-data-mining communities, and QWK / CORN / rank-consistent heads come from computer-vision / ordinal-regression communities. Nobody has bridged the two for this task. That is a genuinely fillable gap.

Option B – Cognitive-level robustness benchmarking in the spirit of BloomAPR

Ma et al. 2025 (BloomAPR) did this for code repair. Nobody has done it for CLO classification. Take the Monash test set and apply *Bloom-preserving perturbations*: verb substitution within a level (e.g. “identify” $\leftrightarrow$ “list”), cross-level adversarial swaps, sentence paraphrase with a held-out LLM. Then measure whether the best 2022 SOTA (BERT) and the 2026 SOTA (QLoRA-Mistral) actually *understand* Bloom or just pattern-match the top bigrams (which your own EDA — “demonstrate understanding”, “critically evaluate”, “apply knowledge” — strongly suggests they do).

Defence: Huber & Niklaus, BloomAPR, and the Findings-EMNLP 2025 cognitive-coverage paper all

argue the field needs *dynamic, perturbation-based* evaluation. You would be first to do this on CLO classification.

Option C — Low-resource / domain-transfer study with QLoRA

Your QLoRA experiments already show 7B $\approx$ DistilBERT on 16K labelled rows. The open question: *how few labelled rows does QLoRA need to match DistilBERT?* Run the same QLoRA at 500, 1K, 2K, 4K, 8K, 16K training rows, plot macro-F1 vs. rows for DistilBERT-full-FT vs. QLoRA-Mistral-7B vs. QLoRA-Qwen2-7B. Then test **cross-institution transfer** — train on Monash, test on a subset of the Yahya / Mohammed question banks, or vice versa. This answers the question Li et al. 2022 explicitly flagged: *does their 21K-row model transfer to other institutions?*

Defence: every 2025 paper on CLO/question classification uses a single dataset. The “do these models transfer?” question is explicitly open in Almatrafi & Johri 2025 (their limitations section) and in Gulwani 2025 (who argue for more data because their 600-sentence models overfit).

Option D — Bloom-aware explainability / abstention

Train the models also to output a calibrated **uncertainty**, and *abstain* on the $\sim 11\%$ of multi-label CLOs rather than forcing a single-label prediction. In your data 10.9% of rows have 2 labels — your current pipeline drops them. A practically useful classifier for educators would say “this outcome spans Apply and Evaluate”. Use DistilBERT with sigmoid heads + conformal prediction / expected calibration error analysis + LLM-based explanation generation (e.g. “This LO is classified as Apply because of the verb ‘implement’ in a procedural context; however, it could also be Analyze if the procedure involves decomposition”).

Defence: Li et al. 2022 themselves found binary-per-class slightly beat multi-class multi-label. Nobody since has asked whether *abstention + explanation* is better than forced single-class prediction. The 2025 RAG-augmented question classification literature (Querla, Toba et al. 2025) is moving this way but has not formalised it.

Option E — Curriculum-alignment application (weakest as sole novelty, strongest as narrative frame)

Build an end-to-end tool: upload a syllabus $\rightarrow$ extract CLOs $\rightarrow$ classify into Bloom levels with your best model $\rightarrow$ flag *structural imbalances* (e.g. “this course has 80% lower-order CLOs — constructive-alignment warning”) $\rightarrow$ suggest revisions using a fine-tuned LLM. Almatrafi & Johri 2025 gestures at this but stops at classification; the 2025 Jackson pedagogy paper in *Student Engagement in Higher Education Journal* explicitly calls for tools like this.

Defence: moves the work from “NLP benchmark” to “educational technology with real deployment” — a different reviewer audience (AIED, LAK, Computers & Education) that values utility over +0.5pp F1.

5.3 Recommended combination

For a **research-paper-worthy** contribution (CAI / AIED / LAK / EDM / COLING-EdNLP), I would combine **A + B** as your primary contributions, use your existing 20-model comparison as background/baseline, and present C as a secondary ablation.

Concretely, the proposed paper title and storyline:

"Rank-Consistent Bloom: Ordinal and Robustness-Aware Classification of Course Learning Outcomes with Open-Source LLMs"

Contribution 1 — We show that the canonical CLO classification setup (nominal CE) silently mis-penalises adjacent-level errors; swapping to ordinal-aware losses (CORN / weighted- κ / class-distance CE) improves QWK by X points without sacrificing macro-F1 on Monash.

Contribution 2 — We introduce a Bloom-perturbation test suite (verb-swap within level, adversarial verb-swap across level, LLM paraphrase) and show that BERT-class and QLoRA-7B models degrade heavily under cross-level perturbation while staying stable under within-level perturbation, evidencing shallow lexical pattern-matching rather than cognitive-level understanding.

Contribution 3 — We release a 20-model calibrated benchmark (classical, deep, zero-shot, few-shot, QLoRA fine-tuned) on the Monash CLO split with a standardised evaluation protocol including QWK and robustness metrics.

That is a paper. What you have right now is the *Contribution 3* half of it.

6. Concrete action plan

Phase 1 ($\approx$ 2 weeks) — Baseline recovery + ordinal pivot

1. Re-run DistilBERT fine-tune and QLoRA-Mistral with (a) weighted cross-entropy, (b) CORN / CORAL ordinal head, © class-distance-weighted CE. Use the `dlordinal` package (Bérchez-Moreno et al. *Neurocomputing* 2025) or `coral-pytorch`.
2. Report QWK + MAE-on-Bloom-levels alongside macro-F1 for every model. Show the ordinal losses win on QWK even when nominal F1 is tied.
3. **Fix the Gemma-9B chat-template bug.** Ship a fixed run. You cannot publish a 20% accuracy result for Gemma without being sure it isn't a template problem; reviewers will ask.

Phase 2 ($\approx$ 3 weeks) — Perturbation benchmark

4. Build 3 perturbation sets: (i) within-level verb swaps using Anderson & Krathwohl's 2001 verb lists, (ii) cross-level adversarial swaps, (iii) LLM paraphrase with GPT-4o-mini or Claude Haiku. Keep a seed-fixed 1,000-row perturbation test set.
5. Run every fine-tuned model on it. Report accuracy drop relative to clean test — this is your “cognitive-robustness” score.

Phase 3 (≈ 1 week) — Writing + re-framing

6. Rewrite Section 1 / abstract around the two new contributions. Demote the 20-model leaderboard to Section 4 or an appendix.
7. Engage directly with Huber & Niklaus 2025, BloomAPR 2025, Almatrafi & Johri 2025, and the MDPI June 2025 Bayesian-ensemble paper in related work — *not* just 2020–2022 sources.
8. Release code + perturbation test set on GitHub. Nobody in this space has released a standardised ordinal + robustness evaluation.

Immediate things to clean up regardless

- The Gemma-2-9B QLoRA collapse is a **bug**, not a result. Your slide 76 correctly identifies chat-template drift. Fix and rerun before publishing; otherwise it will be the *only* thing a reviewer remembers.
- The 10-fold CV for LSTM shows three folds near 24% — this is an initialisation/seed issue, not a model issue. Fix with seed sweeps + early stopping monitoring, or report median-of-5 instead of mean.
- Your few-shot bar charts (slides 53–63) would be more credible with confidence intervals from 3–5 runs with different exemplar orderings. LLM few-shot is exemplar-order-sensitive (this itself is a known 2024 result).

7. Bottom line

You have built a very solid benchmarking infrastructure. What you have *not* built is a research contribution distinguishable from the 2025 literature. The good news: turning this into a contribution does not require throwing anything away. It requires asking a more specific question — *how well do these models respect the ordinal, hierarchical nature of Bloom, and what happens when we perturb the surface verbs that they clearly depend on?* — and using your existing 20-model zoo as the backbone for answering it. That is a paper. Your current slide deck is the appendix to that paper.

8. Key 2025–2026 references to cite

1. Huber, T. & Niklaus, C. (2025). *LLMs meet Bloom's Taxonomy: A Cognitive View on Large Language Model Evaluations*. COLING 2025, pp. 5211–5246.
2. Almatrafi, O. & Johri, A. (2025). *Leveraging generative AI for course learning outcome categorization using Bloom's taxonomy*. Computers and Education: Artificial Intelligence, 8, 100404.
3. Ma, Y. et al. (2025). *BloomAPR: A Bloom's Taxonomy-based Framework for Assessing LLM-Powered APR Solutions*. arXiv:2509.25465.
4. Zoumpoulidi, M.-E., Paraskevopoulos, G., Potamianos, A. (2025). *BloomWise: Enhancing Problem-Solving Capabilities of LLMs using Bloom's-Taxonomy-Inspired Prompts*. MathNLP 2025, pp. 34–49.
5. Gulwani, D. et al. (2025). *Automated Analysis of Learning Outcomes and Exam Questions Based on Bloom's Taxonomy*. arXiv:2511.10903.
6. *Towards Smarter Assessments: Enhancing Bloom's Taxonomy Classification with a Bayesian-Optimized Ensemble Model*. MDPI Electronics 14(12):2312, June 2025.
7. *Do Existing Benchmarks Accurately Evaluate LLMs? From Remembering to Metacognition*. Findings of EMNLP 2025.
8. *Mechanistic Interpretability of Cognitive Complexity in LLMs via Linear Probing using Bloom's Taxonomy*. arXiv:2602.17229, Feb 2026.
9. *Automated Benchmark Generation from Domain Guidelines Informed by Bloom's Taxonomy*. arXiv:2601.20253, Jan 2026.
10. Polat, G. et al. (2025). *Class distance weighted cross entropy loss for classification of disease severity*. Expert Systems with Applications 269, 126372.
11. Bérchez-Moreno, F. et al. (2025). *dlordinal: A Python package for deep ordinal classification*. Neurocomputing 622, 129305.
12. *Clinically Aware Learning: Ordinal Loss Improves Medical Image Classifiers*. Journal of Clinical Medicine 15(1):365, Jan 2026.
13. Toba, H., Yudha et al. (2025). *Zero-shot / few-shot / CoT prompting across LLMs for Bloom alignment — cited in QuerIA*, Expert Systems with Applications (Nov 2025).
14. Li, Y., Rakovic, M., Poh, B.X., Gašević, D., Chen, G. (2022). *Automatic Classification of Learning Objectives Based on Bloom's Taxonomy*. EDM 2022, pp. 530–537 — **the dataset origin; cite as anchor**.

This analysis is honest, not flattering. If you want me to draft any of the new-contribution sections (ordinal loss methodology, perturbation benchmark design, revised abstract), or to produce a side-by-side numerical comparison table with the MDPI 2025 ensemble paper's exact scores, I can do that in a follow-up.

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2.3. The questions you answer have already been answered in 2025. Rapid-fire summary of what 2025 literature already establishes, which your experiments largely re-demonstrate:

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3.1 Direct comparators on CLO / learning-objective classification

Study	Dataset	Best model	Best score
Li, Rakovic, Poh, Gašević, Chen — EDM 2022 (<i>Monash dataset origin</i>)	21,380 Monash CLOs	BERT (binary per-class)	Cohen's $\kappa \leq \mathbf{0.93}$, F1 $\leq \mathbf{0.95}$
Almatrafi & Johri — <i>Computers & Education: AI</i> , 2025	1,000 annotated LOs	BERT fine-tuned (beat every GPT-4 prompt strategy)	not directly comparable; ran CoT, rhetorical-situation, multiple-binary prompts
Ghannam et al. — MDPI <i>Electronics</i> , June 2025 (Bayesian-optimized ensemble, TF-IDF +	Yahya question bank	Soft-vote RF+SVM+GB+XGB, Bayesian-tuned	Accuracy >93.5% (explicitly claims to break the prior ceiling)

Study	Dataset	Best model	Best score
DistilBERT features)			
Gulwani & Kothari — arXiv 2511.10903, Nov 2025	600 sentences, 6 Bloom levels	SVM + synonym aug	94% accuracy (but on 600 sentences, not 18K)
Your study (2026)	Monash 18,773	DistilBERT / QLoRA- Mistral	macro-F1 0.918 / 0.921 , acc 92.7 / 93.2%

Observation: your macro-F1 on the full Monash split is **essentially identical** to what Li et al. already reported in 2022 with vanilla BERT. You did not break the ceiling; you reached it again with more models.

3.2 Adjacent benchmarks with Bloom as the organising axis (2025–2026)

These are the papers your related-work section is going to have to engage with, and which define where the field has moved since 2022:

- **Huber & Niklaus — COLING 2025** (“LLMs meet Bloom’s Taxonomy”): shifted the question from “classify LOs” to “do LLM benchmarks cover the Bloom hierarchy?” Their finding — Evaluate/Create are severely under-represented in MMLU, BBH, AGIEval — is the most-cited 2025 Bloom-LLM paper and will be in every related-work section from now on.
- **Ma et al. — BloomAPR, arXiv Sep 2025**: uses Bloom levels as a *dynamic* benchmark axis for LLM-powered program repair; explicitly shows performance drops across Remember → Analyze (81.57% → 13.46%).
- **Zhou et al. — MedBloomEval, 2025**: Bloom levels as a reasoning-depth axis for medical LLMs.
- **Zoumpoulidi et al. — BloomWise, MathNLP 2025**: uses Bloom levels as a *prompting scaffold* for math reasoning, with an early-stopping criterion when consecutive levels agree.
- **Huber & Niklaus’s linear-probing follow-up (arXiv 2602.17229, Feb 2026)**: mechanistically probes transformer layers for Bloom-level information — finds errors are structured, adjacent-level confusions dominate.
- **"Automated Benchmark Generation from Domain Guidelines" (arXiv 2601.20253, Jan 2026)**: auto-generates Bloom-aligned MCQs from expert guidelines for teaching/dietetics/caregiving; finds LLMs sometimes do *better* on Analyze than on Remember — the opposite of what Huber & Niklaus saw.
- **"Do Existing Benchmarks Accurately Evaluate LLMs?" — Findings of EMNLP 2025** (arXiv 2511): quantifies cognitive-level coverage in 20+ LLM benchmarks using Shannon–Wiener, Simpson, JSD, Gini over Bloom levels.

The field has moved from “can we classify LOs into Bloom levels” (~2020–2022, where you are) to “what can Bloom tell us about LLMs” (2025–2026, where the frontier is). You are asking a 2022 question in 2026.

3.3 Where your numbers do differ from the literature (real findings worth rescuing)

Don't throw everything out. Three genuinely interesting signals in your results that are *not* already in the 2025 literature:

- 1. **Qwen2.5-7B beats Gemma-2-9B in zero/few-shot** but Gemma-2-9B beats Qwen2-7B on few-shot accuracy (67.5% → 72% for Gemma vs. 61% → 75% for Qwen-2.5). This *parameter-count-vs-few-shot-gain* interaction, at 7–9B scale, is not in the 2025 literature I can find.
- 2. **Fine-tuned 7B Mistral (0.921 macro-F1) ≈ fine-tuned DistilBERT (0.918)** on the same split with QLoRA. Almatrafi & Johri 2025 found BERT beats GPT-4 prompt strategies, but they never fine-tuned a 7B open model. Your observation — that QLoRA at ~40M trainable params closes the gap — is a genuinely fresh data point for the CLO task specifically.
- 3. **Few-shot hurts Qwen2.5-1.5B (45.7% → 30.5%)** — a *negative* interaction between exemplars and very small models, in the same family that benefits at 7B. This is the kind of result that *small-model researchers* would actually read.

These are the seeds. Everything else is either a leaderboard or confirmation.

4. Direct conflicts between your results and recent literature

Claim in 2025 lit	Your result	Conflict or not?
Huber & Niklaus 2025: LLMs do better on lower Bloom levels	Your Qwen2.5-7B <i>gains most</i> on Remember (rarest class) in few-shot (+23pp)	Nuance, not conflict. Their claim is about raw ability; yours is about exemplar-induced gains. Worth articulating.
Gulwani 2025 (arXiv 2511.10903): OpenAI/ Gemini ≈ 0.72–0.73 acc zero-shot, beats open LLMs	Qwen2.5-7B hits 0.61 ZS, 0.75 FS; Gemma-2-9B 0.675 ZS, 0.72 FS	No real conflict — different dataset (600 sentences vs 18K), but you could claim that <i>at 7–9B</i> , open models now match GPT-4o-mini zero-shot, which is a useful point.

Claim in 2025 lit	Your result	Conflict or not?
MDPI 2025 Bayesian ensemble: >93.5% is achievable	Your best is 93.2%	You are below. They beat you.
Almatrafi & Johri 2025: fine-tuned BERT > best-prompted GPT-4	DistilBERT 0.918 ≈ QLoRA-Mistral 0.921 > any of your prompted LLMs (best FS ≈ 0.75 acc)	Confirms their finding, does not challenge it.
BloomAPR 2025: LLM performance cliffs at Analyze layer (~43% Apply → ~13% Analyze on perturbed data)	DistilBERT confusion: Analyze recall 0.91, Evaluate 0.91 — no cliff	Genuine conflict on surface , but they evaluate reasoning, not text classification. Don't overclaim.

5. Your three questions, answered directly

5.1 “How different are my results from previous studies?”

Short answer: barely. On the Monash dataset you match Li et al. 2022 (BERT $\kappa=0.93$, $F1=0.95$ → you get macro-F1 0.918–0.921). On the broader Bloom-NLP literature, you are in the same 72–94% accuracy band as every 2024–2025 study, with no clear win. The three *genuinely differentiated* findings are listed in §3.3. If you write the paper as it stands, a reviewer will write: “*The authors confirm known results with a larger model zoo. Contribution is incremental.*”

5.2 “What novelty / new methodology should I bring?”

Here are **five candidate contributions**, ranked by novelty-defensibility-cost trade-off. Each cites why the 2024–2026 literature has left it open.

Option A — Hierarchy-aware / ordinal classification for Bloom ★ strongest defense

Bloom’s six levels are *ordinal* (Remember < Understand < Apply < Analyze < Evaluate < Create), but every single paper I found — Li et al. 2022, Almatrafi & Johri 2025, MDPI *Electronics* 2025, Gulwani 2025, your study — trains with **nominal cross-entropy**. Misclassifying Create as Remember is penalised the same as misclassifying Create as Evaluate. This is obviously wrong pedagogically and

wrong statistically.

Why it's defensible (2025–2026 evidence):

- The BloomBERT 2025 GitHub README explicitly calls this out as open future work (“ordinal regression formulation could enforce such constraints”).
- Huber & Niklaus’s Feb 2026 linear-probing paper (arXiv 2602.17229) shows empirically that “*misclassifications predominantly occur between adjacent Bloom levels*” — the error structure is ordinal, but the loss is not.
- The ordinal-loss literature is mature (CORAL, CORN, weighted- κ loss, class-distance-weighted CE — Polat et al. 2025 *Expert Systems with Applications*; Shi, Cao & Raschka 2023; Bérchez-Moreno et al. *Neurocomputing* 2025 “dlordinal” Python package).
- In medical imaging 2025–2026, swapping CE for ordinal-aware loss gives consistent macro-F1 gains on BI-RADS and lumbar CT (both *Diagnostics / Journal of Clinical Medicine*, late-2025 MDPI).

What you'd actually do: fine-tune DistilBERT and/or Mistral-7B QLoRA with (a) weighted- κ loss, (b) CORAL-style rank-consistent output heads, © class-distance-weighted cross-entropy. Report not just macro-F1 but **quadratic weighted kappa (QWK)** and **mean absolute error in Bloom levels** — metrics the current Bloom literature *does not report*. Show that models look ~equivalent on nominal F1 but differ meaningfully on QWK.

Why nobody has done this for Bloom specifically: because CLO classification grew out of educational-data-mining communities, and QWK / CORN / rank-consistent heads come from computer-vision / ordinal-regression communities. Nobody has bridged the two for this task. That is a genuinely fillable gap.

Option B — Cognitive-level robustness benchmarking in the spirit of BloomAPR

Ma et al. 2025 (BloomAPR) did this for code repair. Nobody has done it for CLO classification. Take the Monash test set and apply *Bloom-preserving perturbations*: verb substitution within a level (e.g. “identify” $\leftrightarrow$ “list”), cross-level adversarial swaps, sentence paraphrase with a held-out LLM. Then measure whether the best 2022 SOTA (BERT) and the 2026 SOTA (QLoRA-Mistral) actually *understand* Bloom or just pattern-match the top bigrams (which your own EDA — “demonstrate understanding”, “critically evaluate”, “apply knowledge” — strongly suggests they do).

Defence: Huber & Niklaus, BloomAPR, and the Findings-EMNLP 2025 cognitive-coverage paper all argue the field needs *dynamic, perturbation-based* evaluation. You would be first to do this on CLO classification.

Option C — Low-resource / domain-transfer study with QLoRA

Your QLoRA experiments already show 7B $\approx$ DistilBERT on 16K labelled rows. The open question:

how few labelled rows does QLoRA need to match DistilBERT? Run the same QLoRA at 500, 1K, 2K, 4K, 8K, 16K training rows, plot macro-F1 vs. rows for DistilBERT-full-FT vs. QLoRA-Mistral-7B vs. QLoRA-Qwen2-7B. Then test **cross-institution transfer** – train on Monash, test on a subset of the Yahya / Mohammed question banks, or vice versa. This answers the question Li et al. 2022 explicitly flagged: *does their 21K-row model transfer to other institutions?*

Defence: every 2025 paper on CLO/question classification uses a single dataset. The “do these models transfer?” question is explicitly open in Almatrafi & Johri 2025 (their limitations section) and in Gulwani 2025 (who argue for more data because their 600-sentence models overfit).

Option D – Bloom-aware explainability / abstention

Train the models also to output a calibrated **uncertainty**, and *abstain* on the ~11% of multi-label CLOs rather than forcing a single-label prediction. In your data 10.9% of rows have 2 labels – your current pipeline drops them. A practically useful classifier for educators would say “this outcome spans Apply and Evaluate”. Use DistilBERT with sigmoid heads + conformal prediction / expected calibration error analysis + LLM-based explanation generation (e.g. “This LO is classified as Apply because of the verb ‘implement’ in a procedural context; however, it could also be Analyze if the procedure involves decomposition”).

Defence: Li et al. 2022 themselves found binary-per-class slightly beat multi-class multi-label. Nobody since has asked whether *abstention + explanation* is better than forced single-class prediction. The 2025 RAG-augmented question classification literature (Querla, Toba et al. 2025) is moving this way but has not formalised it.

Option E – Curriculum-alignment application (weakest as sole novelty, strongest as narrative frame)

Build an end-to-end tool: upload a syllabus → extract CLOs → classify into Bloom levels with your best model → flag *structural imbalances* (e.g. “this course has 80% lower-order CLOs – constructive-alignment warning”) → suggest revisions using a fine-tuned LLM. Almatrafi & Johri 2025 gestures at this but stops at classification; the 2025 Jackson pedagogy paper in *Student Engagement in Higher Education Journal* explicitly calls for tools like this.

Defence: moves the work from “NLP benchmark” to “educational technology with real deployment” – a different reviewer audience (AIED, LAK, Computers & Education) that values utility over +0.5pp F1.

5.3 Recommended combination

For a **research-paper-worthy** contribution (CAI / AIED / LAK / EDM / COLING-EdNLP), I would combine **A + B** as your primary contributions, use your existing 20-model comparison as

background/baseline, and present C as a secondary ablation.

Concretely, the proposed paper title and storyline:

"Rank-Consistent Bloom: Ordinal and Robustness-Aware Classification of Course Learning Outcomes with Open-Source LLMs"

Contribution 1 — We show that the canonical CLO classification setup (nominal CE) silently mis-penalises adjacent-level errors; swapping to ordinal-aware losses (CORN / weighted- κ / class-distance CE) improves QWK by X points without sacrificing macro-F1 on Monash.

Contribution 2 — We introduce a Bloom-perturbation test suite (verb-swap within level, adversarial verb-swap across level, LLM paraphrase) and show that BERT-class and QLoRA-7B models degrade heavily under cross-level perturbation while staying stable under within-level perturbation, evidencing shallow lexical pattern-matching rather than cognitive-level understanding.

Contribution 3 — We release a 20-model calibrated benchmark (classical, deep, zero-shot, few-shot, QLoRA fine-tuned) on the Monash CLO split with a standardised evaluation protocol including QWK and robustness metrics.

That is a paper. What you have right now is the *Contribution 3* half of it.

6. Concrete action plan

Phase 1 ($\approx$ 2 weeks) — Baseline recovery + ordinal pivot

1. Re-run DistilBERT fine-tune and QLoRA-Mistral with (a) weighted cross-entropy, (b) CORN / CORAL ordinal head, © class-distance-weighted CE. Use the `dlordinal` package (Bérchez-Moreno et al. *Neurocomputing* 2025) or `coral-pytorch`.
2. Report QWK + MAE-on-Bloom-levels alongside macro-F1 for every model. Show the ordinal losses win on QWK even when nominal F1 is tied.
3. **Fix the Gemma-9B chat-template bug.** Ship a fixed run. You cannot publish a 20% accuracy result for Gemma without being sure it isn't a template problem; reviewers will ask.

Phase 2 ($\approx$ 3 weeks) — Perturbation benchmark

4. Build 3 perturbation sets: (i) within-level verb swaps using Anderson & Krathwohl's 2001 verb lists, (ii) cross-level adversarial swaps, (iii) LLM paraphrase with GPT-4o-mini or Claude Haiku. Keep a seed-fixed 1,000-row perturbation test set.

5. Run every fine-tuned model on it. Report accuracy drop relative to clean test — this is your “cognitive-robustness” score.

Phase 3 (≈ 1 week) — Writing + re-framing

6. Rewrite Section 1 / abstract around the two new contributions. Demote the 20-model leaderboard to Section 4 or an appendix.
7. Engage directly with Huber & Niklaus 2025, BloomAPR 2025, Almatrafi & Johri 2025, and the MDPI June 2025 Bayesian-ensemble paper in related work — *not* just 2020–2022 sources.
8. Release code + perturbation test set on GitHub. Nobody in this space has released a standardised ordinal + robustness evaluation.

Immediate things to clean up regardless

- The Gemma-2-9B QLoRA collapse is a **bug**, not a result. Your slide 76 correctly identifies chat-template drift. Fix and rerun before publishing; otherwise it will be the *only* thing a reviewer remembers.
- The 10-fold CV for LSTM shows three folds near 24% — this is an initialisation/seed issue, not a model issue. Fix with seed sweeps + early stopping monitoring, or report median-of-5 instead of mean.
- Your few-shot bar charts (slides 53–63) would be more credible with confidence intervals from 3–5 runs with different exemplar orderings. LLM few-shot is exemplar-order-sensitive (this itself is a known 2024 result).

7. Bottom line

You have built a very solid benchmarking infrastructure. What you have *not* built is a research contribution distinguishable from the 2025 literature. The good news: turning this into a contribution does not require throwing anything away. It requires asking a more specific question — *how well do these models respect the ordinal, hierarchical nature of Bloom, and what happens when we perturb the surface verbs that they clearly depend on?* — and using your existing 20-model zoo as the backbone for answering it. That is a paper. Your current slide deck is the appendix to that paper.

8. Key 2025–2026 references to cite

1. Huber, T. & Niklaus, C. (2025). *LLMs meet Bloom's Taxonomy: A Cognitive View on Large Language Model Evaluations*. COLING 2025, pp. 5211–5246.
2. Almatrafi, O. & Johri, A. (2025). *Leveraging generative AI for course learning outcome categorization using Bloom's taxonomy*. *Computers and Education: Artificial Intelligence*, 8, 100404.
3. Ma, Y. et al. (2025). *BloomAPR: A Bloom's Taxonomy-based Framework for Assessing LLM-Powered APR Solutions*. arXiv:2509.25465.
4. Zoumpoulidi, M.-E., Paraskevopoulos, G., Potamianos, A. (2025). *BloomWise: Enhancing Problem-Solving Capabilities of LLMs using Bloom's-Taxonomy-Inspired Prompts*. MathNLP 2025, pp. 34–49.
5. Gulwani, D. et al. (2025). *Automated Analysis of Learning Outcomes and Exam Questions Based on Bloom's Taxonomy*. arXiv:2511.10903.
6. *Towards Smarter Assessments: Enhancing Bloom's Taxonomy Classification with a Bayesian-Optimized Ensemble Model*. MDPI Electronics 14(12):2312, June 2025.
7. *Do Existing Benchmarks Accurately Evaluate LLMs? From Remembering to Metacognition*. Findings of EMNLP 2025.
8. *Mechanistic Interpretability of Cognitive Complexity in LLMs via Linear Probing using Bloom's Taxonomy*. arXiv:2602.17229, Feb 2026.
9. *Automated Benchmark Generation from Domain Guidelines Informed by Bloom's Taxonomy*. arXiv:2601.20253, Jan 2026.
10. Polat, G. et al. (2025). *Class distance weighted cross entropy loss for classification of disease severity*. *Expert Systems with Applications* 269, 126372.
11. Bérchez-Moreno, F. et al. (2025). *dlordinal: A Python package for deep ordinal classification*. *Neurocomputing* 622, 129305.
12. *Clinically Aware Learning: Ordinal Loss Improves Medical Image Classifiers*. *Journal of Clinical Medicine* 15(1):365, Jan 2026.
13. Toba, H., Yudha et al. (2025). *Zero-shot / few-shot / CoT prompting across LLMs for Bloom alignment — cited in QuerIA, Expert Systems with Applications (Nov 2025)*.
14. Li, Y., Rakovic, M., Poh, B.X., Gašević, D., Chen, G. (2022). *Automatic Classification of Learning Objectives Based on Bloom's Taxonomy*. EDM 2022, pp. 530–537 — **the dataset origin; cite as anchor**.

This analysis is honest, not flattering. If you want me to draft any of the new-contribution sections (ordinal loss methodology, perturbation benchmark design, revised abstract), or to produce a side-by-side numerical comparison table with the MDPI 2025 ensemble paper's exact scores, I can do that in a follow-up.